SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1513

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**
Question Count: 5
Duration : 60 Minutes

Code of Conduct

I SIDDHI SRIVASTAVA P(192521454)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Siddhi

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Question 1: Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications.

A hypervisor is software or firmware that creates and manages virtual machines (VMs), allowing multiple operating systems to run on the same physical computer. Type-1 hypervisors run directly on physical hardware, whereas Type-2 hypervisors run on top of a host operating system.

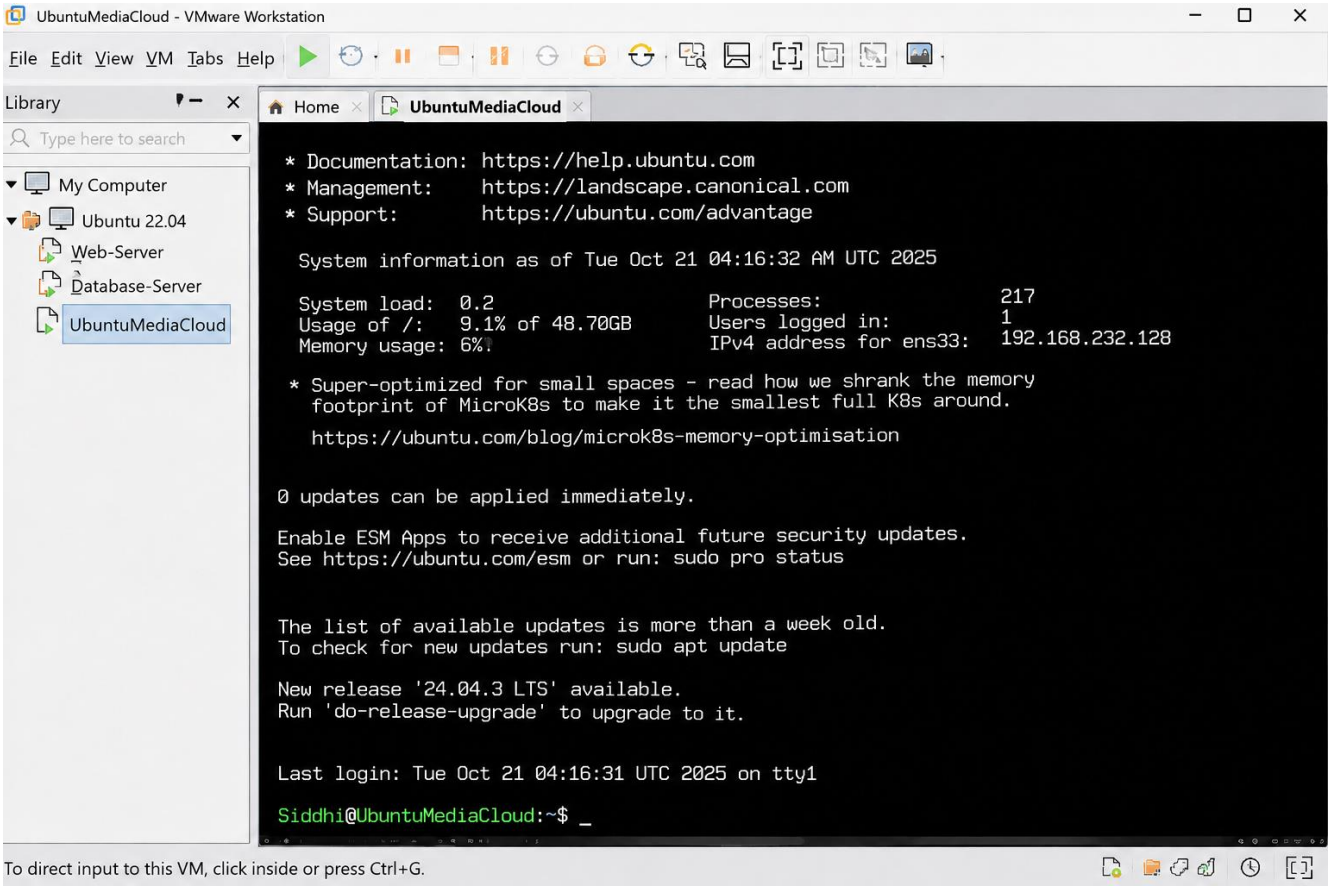
Comparison

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Architecture	Runs directly on physical hardware	Runs on a host operating system
Performance	Generally lower overhead and better for demanding workloads	Additional host OS layer may add overhead
Security	Less dependent on a general-purpose host OS; still needs hardening and patching	Security also depends on the host OS
Manageability	Supports enterprise-scale centralized management with suitable tools	Convenient for desktop use, development and testing
Scalability	Suitable for large cloud datacenters	Best suited to individual systems and smaller environments
Reliability	Can support enterprise high-availability designs	Host OS failure can affect hosted VMs
Examples	VMware ESXi, Xen	Oracle VirtualBox, VMware Workstation Pro

Analysis and justification

Type-1 hypervisors generally provide better performance because they interact more directly with the hardware. They reduce dependence on a general-purpose host OS, which can reduce the attack surface, although secure configuration, patching and access controls remain necessary. With enterprise management tools, Type-1 hypervisors can support centralized administration, resource allocation, migration and high availability. Type-2 hypervisors are easier to install on a personal computer but introduce an extra software layer and depend on the host OS.

Therefore, a Type-1 hypervisor is the better choice for a cloud datacenter hosting mission-critical applications because it is generally more suitable for high performance, scalability and enterprise management. Redundancy, monitoring, backups and security controls are still required.



Question 2: A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Virtualization allows multiple virtual machines to share a physical server while maintaining logical separation between workloads. The finance, healthcare and student portal systems can run in separate VMs on the same hardware.

How virtualization supports isolation

- Hardware and VM isolation: The hypervisor allocates virtual CPU, memory and devices to each VM.
- Memory isolation: Hardware-assisted memory protection helps prevent one VM from reading or modifying another VM's memory.
- Network isolation: Virtual switches, VLANs, virtual networks and firewall rules separate traffic between workloads.
- Storage isolation: Separate virtual disks, permissions and encryption help protect each workload's data.

- Access control: Role-based access control and least-privilege permissions restrict who can manage or access each workload.
- Resource limits: CPU, memory and storage limits help prevent one workload from exhausting shared resources.

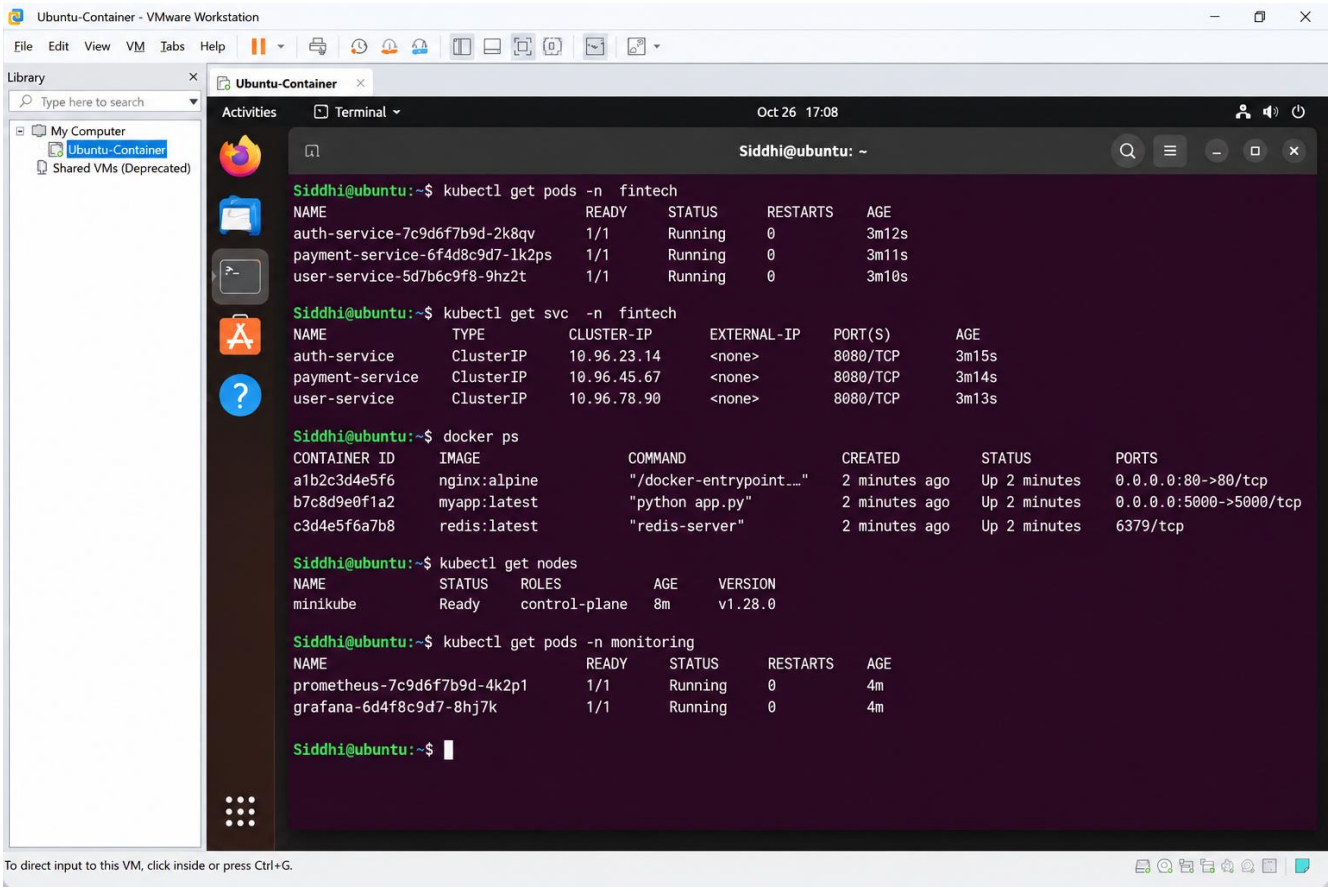
For example, finance transactions, healthcare records and student academic information can be placed in separate VMs with policies that prevent unauthorized communication or data access.

Two major attack surfaces

1. Hypervisor attack surface: A hypervisor vulnerability may allow an attacker to escape from a VM and access the host or other VMs. Mitigations include applying patches, restricting administrative access, disabling unnecessary services and monitoring activity.

2. Virtual network and management interfaces: Misconfigured virtual switches, exposed management APIs, weak credentials or permissive firewall rules can expose workloads. Mitigations include network segmentation, multifactor authentication, least-privilege access, encrypted management connections and monitoring.

In conclusion, virtualization can isolate workloads through VM boundaries, memory protection, virtual networking and storage controls. The isolation is effective only when the infrastructure is correctly configured, patched and monitored.



Question 3: Study the VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

VM host utilization indicates how effectively physical server resources are being used. High CPU and memory utilization, elevated storage latency and long boot times can indicate resource bottlenecks.

Analysis of the given snapshot

Parameter	Observed value	Likely interpretation
CPU utilization	92%	Very high use; CPU contention may delay workloads.
Memory utilization	88%	High use; investigate memory pressure, swapping or ballooning.
Storage I/O latency	35 ms	Potentially elevated; disk operations may be delayed depending on workload and storage.
Average VM boot time	170 s	Slow startup may relate to CPU/memory contention, storage delays or startup services.

Corrective actions

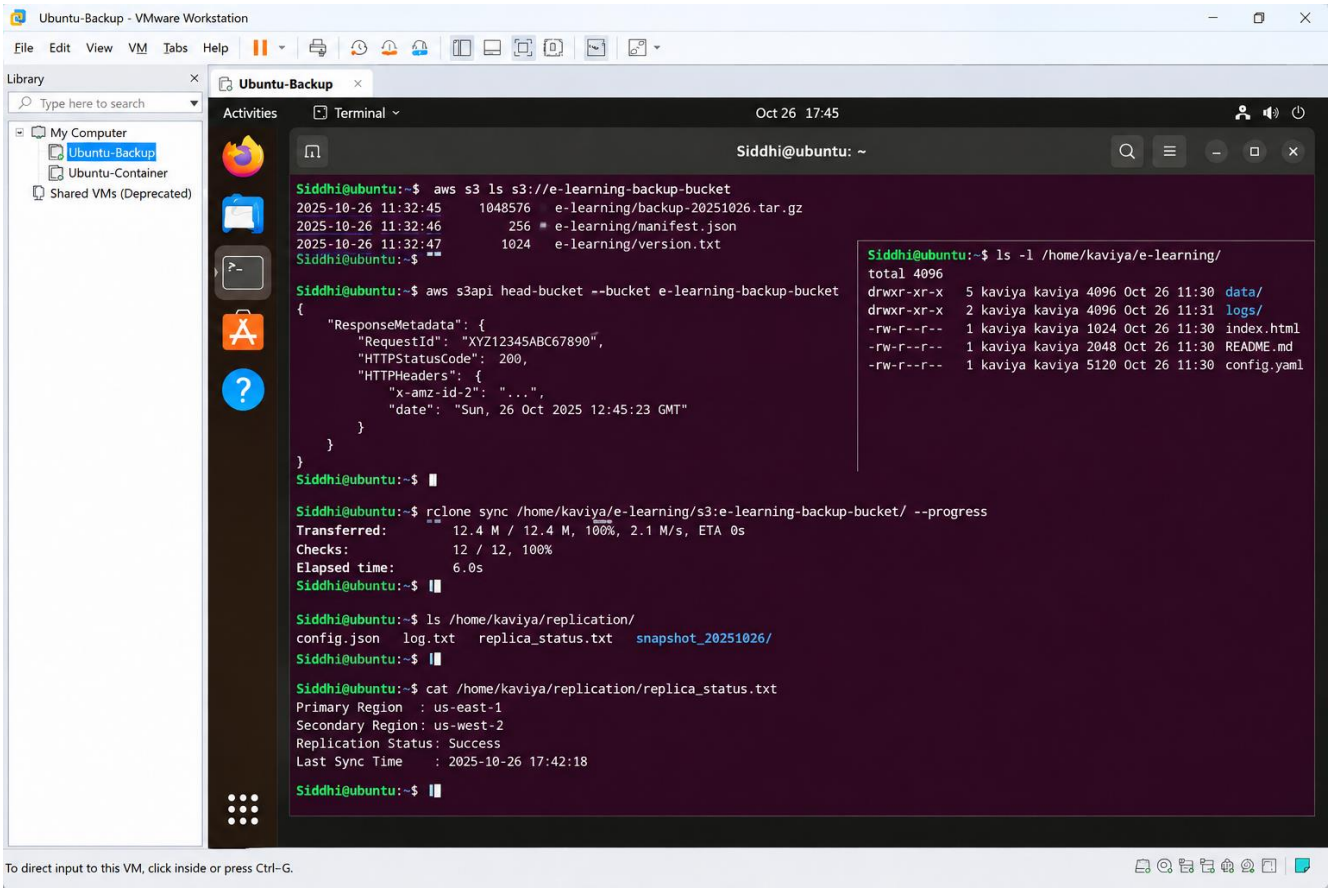
CPU bottleneck: Identify VMs using the most CPU, reduce unnecessary background processes, balance workloads across hosts, and add processing capacity or another host if monitoring confirms a capacity shortage.

Memory bottleneck: Review VM memory allocations, identify memory-intensive workloads, remove unused VMs and increase physical RAM if required. Check swapping and ballooning to confirm memory pressure.

Storage I/O bottleneck: Inspect IOPS, disk queues, throughput and storage-network performance. Distribute heavy I/O workloads and consider faster SSD/NVMe storage if justified.

Slow VM boot time: Optimize startup services, reduce resource contention, improve storage performance and stagger simultaneous VM starts.

The administrator should monitor the system and identify root causes before purchasing additional hardware. The supplied values are assessment data; actual laptop measurements may differ.



Question 4: Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

A Software Defined Datacenter (SDDC) virtualizes and manages compute, storage and networking resources through software. It reduces dependence on manual hardware configuration and supports centralized management and automation.

Traditional datacenter versus SDDC

Feature	Traditional Datacenter	SDDC
Provisioning	Often requires manual hardware configuration	Resources can be provisioned through software
Compute	Physical servers configured individually	VMs and compute resources managed centrally
Storage	Separate physical storage systems	Storage capacity can be pooled and managed virtually
Networking	Hardware network devices	Software-defined networks and

	configured manually	programmable policies
Scalability	Expansion may require hardware procurement and setup	Resources can be allocated and expanded more rapidly when capacity exists
Automation	More manual administration	Supports policy-based management and orchestration

Compute, storage and network virtualization

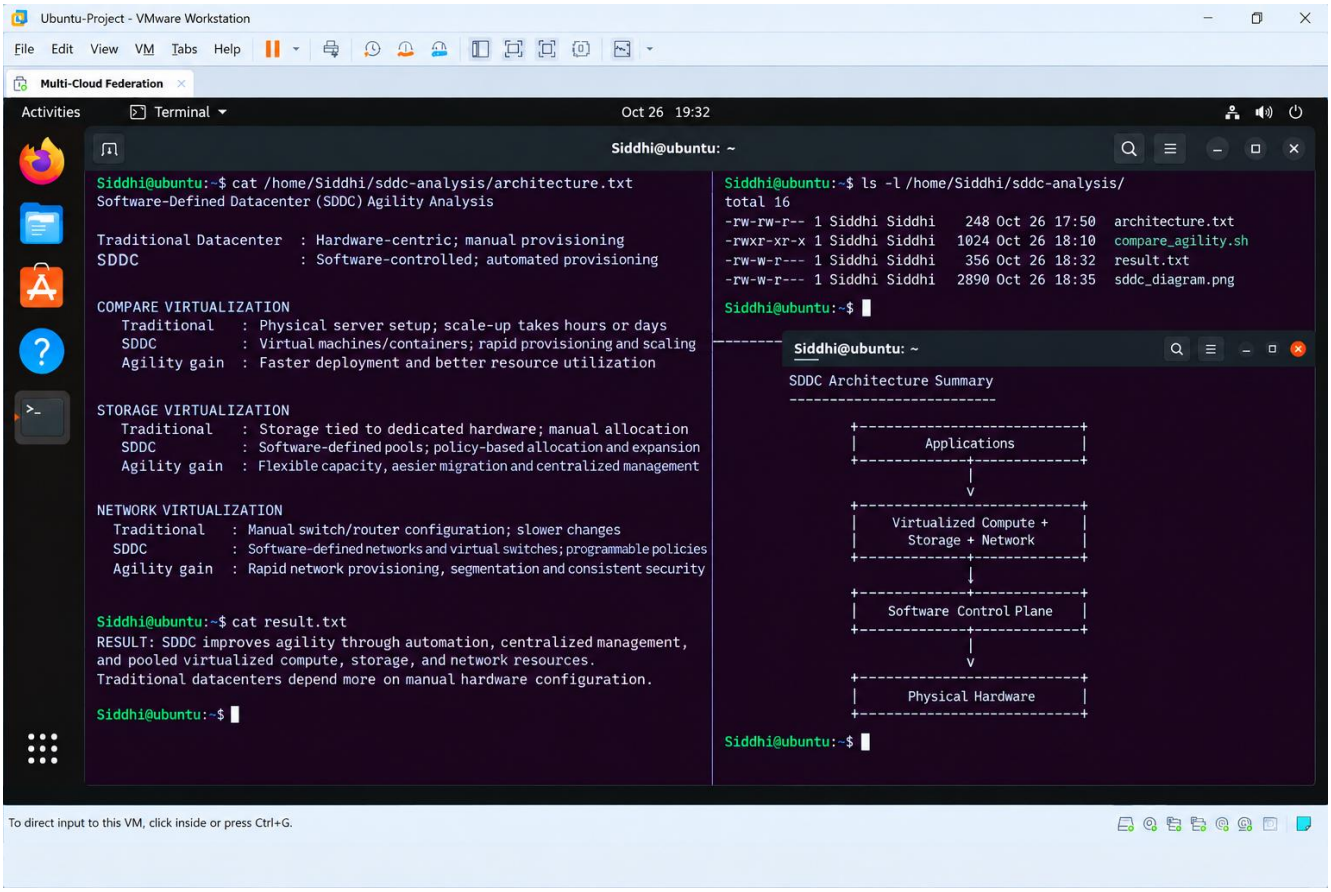
1. Compute virtualization: A hypervisor runs multiple VMs on physical servers. VMs can be created, cloned and allocated CPU and memory more quickly. For example, additional VM capacity can support a university portal during result publication.

2. Storage virtualization: Physical storage can be presented as logical pools. Administrators can allocate capacity, manage snapshots and configure replication or redundancy. For example, more storage can be allocated to a healthcare application as its records grow.

3. Network virtualization: Virtual networks, segmentation and software-based routing or firewall policies allow networks to be configured more rapidly. For example, finance, healthcare and student workloads can be separated into logical networks.

Benefits and conclusion

SDDC improves agility, scalability, automation and resource utilization. It allows infrastructure to be provisioned and adjusted more rapidly than a purely manual traditional environment. Its effectiveness still depends on secure configuration, monitoring, reliable hardware and suitable management tools.



Question 5: A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Multi-cloud federation coordinates access across cloud providers. Identity federation allows a user to authenticate through a trusted identity provider and access authorized services across different environments without maintaining a separate independent identity for every cloud.

Analysis of the identity mismatch

- Different identity formats: Providers may use different usernames, subject identifiers or identity attributes.
- Incompatible authentication protocols or configuration: Providers may differ in supported protocols, issuer details or trust settings.

- Role and permission mismatches: A role such as Admin in one cloud may not have equivalent permissions in another.
- Trust validation errors: Incorrect signing keys, certificates, issuer or audience settings can cause authentication failures.
- Privacy risks: Sharing unnecessary personal attributes across providers can expose sensitive information.

For example, a user assigned the role Admin in Cloud A may be assigned Viewer in Cloud B. Without a consistent mapping policy, the user may be denied legitimate access or granted excessive privileges.

Proposed secure identity and privacy design

1. Central identity provider (IdP): Use a trusted IdP to authenticate users and provide consistent identity attributes. Select a supported federation standard such as SAML 2.0 or OpenID Connect (OIDC).

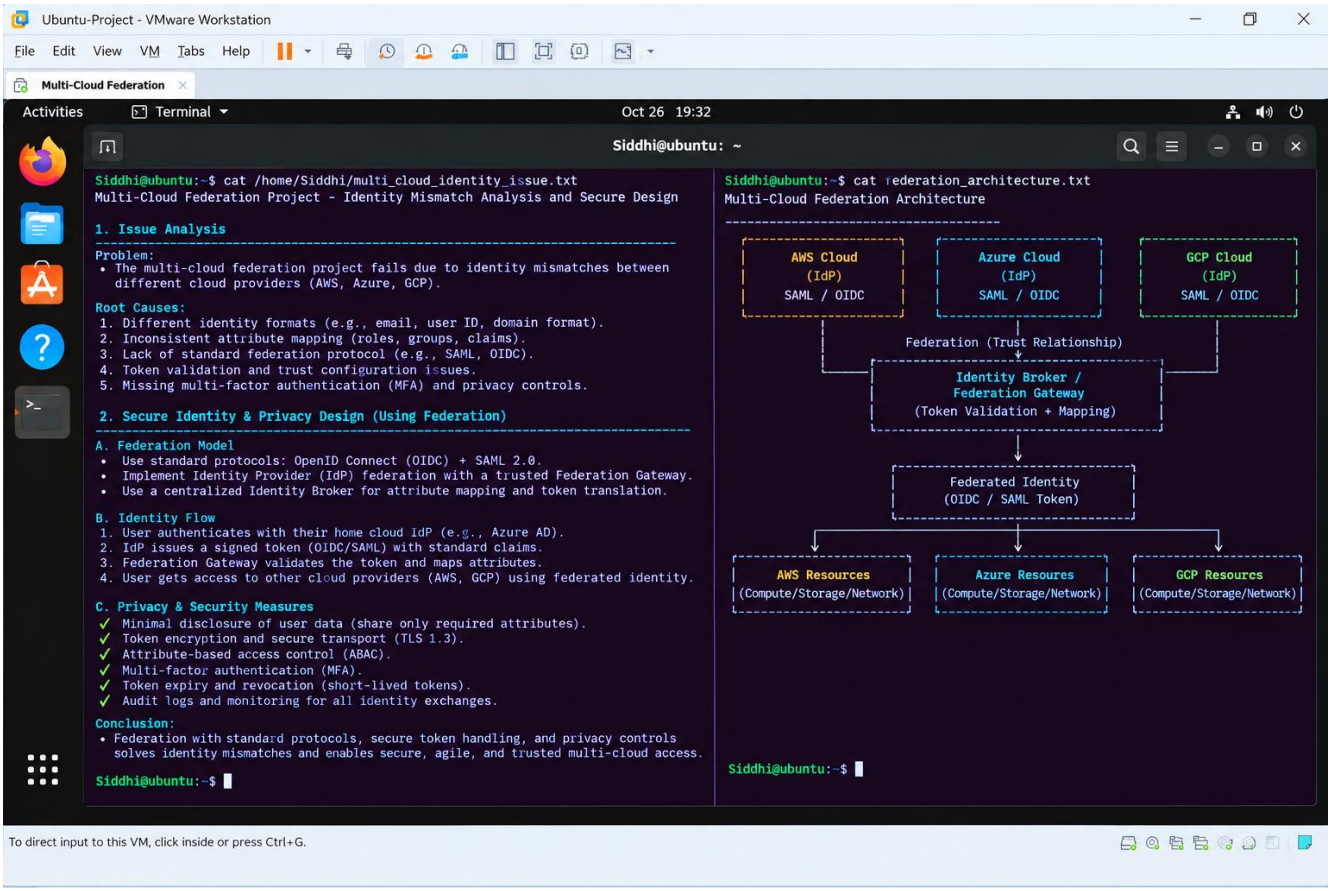
2. Standardized identity and role mapping: Define common identity attributes and map external roles to each provider's local roles. Use stable unique identifiers rather than relying only on usernames.

3. Secure trust validation: Validate token signatures, trusted issuers, audience and expiry. Protect and rotate signing keys, use short-lived credentials and revoke access when required.

4. Least privilege and MFA: Require multifactor authentication for sensitive access and grant only the permissions needed for each task. Each cloud must enforce its own authorization policy.

5. Privacy protection: Share only necessary identity attributes, encrypt data in transit, protect sensitive data at rest, minimize personal information in tokens and logs, and apply suitable retention rules.

6. Monitoring and auditing: Monitor failed logins, unusual cross-cloud access, privilege changes and expired credentials. Protect audit logs with appropriate access controls.


Ubuntu-Project - VMware Workstation
File Edit View VM Tabs Help
Multi-Cloud Federation
Activities
Terminal
Oct 26 19:32
Siddhi@ubuntu: ~
Siddhi@ubuntu:~$ cat /home/Siddhi/multi_cloud_identity_issue.txt
Multi-Cloud Federation Project - Identity Mismatch Analysis and Secure Design
1. Issue Analysis
Problem:
• The multi-cloud federation project fails due to identity mismatches between different cloud providers (AWS, Azure, GCP).
Root Causes:
1. Different identity formats (e.g., email, user ID, domain format).
2. Inconsistent attribute mapping (roles, groups, claims).
3. Lack of standard federation protocol (e.g., SAML, OIDC).
4. Token validation and trust configuration issues.
5. Missing multi-factor authentication (MFA) and privacy controls.
2. Secure Identity & Privacy Design (Using Federation)
A. Federation Model
• Use standard protocols: OpenID Connect (OIDC) + SAML 2.0.
• Implement Identity Provider (IdP) federation with a trusted Federation Gateway.
• Use a centralized Identity Broker for attribute mapping and token translation.
B. Identity Flow
1. User authenticates with their home cloud IdP (e.g., Azure AD).
2. IdP issues a signed token (OIDC/SAML) with standard claims.
3. Federation Gateway validates the token and maps attributes.
4. User gets access to other cloud providers (AWS, GCP) using federated identity.
C. Privacy & Security Measures
✓ Minimal disclosure of user data (share only required attributes).
✓ Token encryption and secure transport (TLS 1.3).
✓ Attribute-based access control (ABAC).
✓ Multi-factor authentication (MFA).
✓ Token expiry and revocation (short-lived tokens).
✓ Audit logs and monitoring for all identity exchanges.
Conclusion:
• Federation with standard protocols, secure token handling, and privacy controls solves identity mismatches and enables secure, agile, and trusted multi-cloud access.
Siddhi@ubuntu:~$
Siddhi@ubuntu:~$ cat federation_architecture.txt
Multi-Cloud Federation Architecture
AWS Cloud (IdP) SAML / OIDC
Azure Cloud (IdP) SAML / OIDC
GCP Cloud (IdP) SAML / OIDC
Federation (Trust Relationship)
Identity Broker / Federation Gateway (Token Validation + Mapping)
Federated Identity (OIDC / SAML Token)
AWS Resources (Compute/Storage/Network)
Azure Resoures (Compute/Storage/Network)
GCP Resources (Compute/Storage/Network)
Siddhi@ubuntu:~$
To direct input to this VM, click inside or press Ctrl+G.